

Section 9: Startup, Adjustments, and Shutdown

A. Startup

General Startup Information

When possible, Johnston burners are factory test fired to ensure that components and systems are functional. All burners supplied on Johnston boilers are test fired as units, to as close to the minimum and maximum firing rates as practical. The fuel air ratio is adjusted as much as possible with the factory conditions. Because each field application is unique due to flue stack configurations, fuel composition variations, elevation, and other variables, the final burner setup and calibration must be conducted after system installation by a qualified burner service technician. Contact your Johnston Boiler Representative or the Johnston Boiler factory service department for scheduling and rates of factory authorized commissioning assistance.

Burners supplied as stand alone items will not be fire tested.

Where the word “boiler” is used, it will be synonymous with chamber or other device to which the burner is applied.

Pre-Startup Checks: Boiler or Applicable Chamber

Note: If your Johnston burner is to be installed on an apparatus not made by Johnston Boiler Company, follow the startup procedure given by the manufacturer of the product.

- ☐ The boiler/chamber is properly installed on a concrete pad, grouted and is level.
- ☐ The flue stack has been installed. A rain cap has been installed if directly connected to the flue gas outlet.
- ☐ The boiler/chamber's piping, blowoff, blowdown, and feedwater systems have all been installed and connected according to the manufacturer's instructions.
- ☐ The boiler/chamber's doors and access openings are fully accessible.

NOTE:

Each Johnston burner is shipped with an order specification sheet, bills of material, piping diagram, wiring diagram, and other reference materials specific to this job. A thorough understanding of, and familiarity with, these documents is required prior to systems installation, commissioning, operation or trouble shooting.

Pre-Startup Checks: Burner

- ☐ The electrical power supply has been run to the control panel. A disconnect has been installed at the control panel, just before the power supply enters the control panel. If the control panel was supplied by others, ensure that it is wired to match the wiring diagrams that came with your burner.
- ☐ All motor rotations have been checked (blower, oil pump, air compressor) against the direction of rotation labeled on the motor. Due to job-specific differences in motor setup, proper motor rotation is labeled by the factory using arrows.
- ☐ There is nothing about the installation that will no longer allow the burner and fuel train(s) to operate properly and be adjusted, as may be necessary.
- ☐ Fuel supply pressures match the specification sheet.
- ☐ Continuity and electrical check of any wiring that was required into the control panel.
- ☐ For oil firing, fuel oil has been circulated and strainers are empty and clean.

Section 9:

WARNING:

Commissioning or adjustment of the Johnston Burner must be made by factory authorized technicians and experienced service technicians to ensure safe and satisfactory operation of the system.

NOTE:

The following sequences assume that a Johnston boiler control panel and burner management system have been supplied with the burner. If these items have been supplied by others, the general sequence and intent will be as follows, but the control system manufacturer will have to be consulted for operational details.

WARNING:

Do not attempt to start up a boiler or burner unless you have read and understand this entire manual. Only qualified and experienced boiler, burner, and controls start-up and service personnel should attempt to start any boiler/burner.

NOTE:

Do not attempt to light, or re-light, the pilot or main burner of any boiler, on any fuel, if the presence of an excess amount of fuel is noted in the fireside of the boiler, whether this is noted through smell, or visible to the eye. Prior to light off, the boiler/chamber must be purged free of any unburned fuel.

CAUTION:

During initial startup, the operator must be on constant alert for conditions such as leaks and electrical malfunctions. The operator should be constantly aware of the positions of manual shutdown valves and switches to ensure fast manual shutdown if necessary.

Startup Procedure: Gas

1. Make sure that all linkages, arms, and connection points are tight in the fuel/air ratio system.
2. Ensure that the burner is in gas firing mode, with the shifter lever positioned for gas firing.
3. Close the main shutoff cock and the supervisor shutoff cock on the main gas train, and the manual shutoff cock on the pilot gas train.
4. Check the gas supply for the required pressure. Make sure that the supply line has been bled. Make sure that a "dirt trap" has been installed in the supply line, just before it connects up to the boiler main gas train.
5. Install the regulated and manifold gas pressure gauges at their proper location in the main gas train. Very often the regulated pressure gauge is a higher range gauge than the manifold pressure gauge.
6. Purge the boiler with the pilot gas valve and main gas train valves closed. Once the purge is done, check visually for a pilot spark. Make sure that the scanner does not sense the spark.
7. Open the pilot gas shutoff cock slowly.
8. Adjust the main gas train regulator to optimize pilot flame (refer to flame safeguard manufacturer for proper signals).
9. With the downstream (second) checking shutoff cock closed, slowly open the upstream (first) manual shutoff valve.

Do not confuse supply or regulated lockup pressures (as the result of closed downstream valves or shutoff cocks) with continuous flow operating pressures. Continuous flow operating pressure will only be accurate when the boiler is operating.

10. When initially starting a boiler, after a pilot flame has been established, slowly open the downstream (second) checking cock as the main gas shutoff valves open.

Section 9:

11. Check low fire regulated gas pressure against the reading shown on the test fire report located in following pages of this manual. The low fire regulated gas pressure should be the same as shown on the test fire report.

CAUTION:

Every potential operating situation or system condition cannot be foreseen or documented. The following instructions should be used as a commissioning guideline by competent and experienced burner service technicians. Specific situations may mandate deviations based on sound judgment and engineering experience. Consult the Johnston Burner factory if questions or problems are present.

Startup Procedure: Oil

1. Check that all linkages are in working order.
2. Ensure that the burner is in oil firing mode, with the shifter lever positioned for oil firing.
3. Check for a manual shutoff valve in the fuel oil supply line, upstream of the fuel oil strainer. Make sure that this shutoff valve is open.
4. Check to see that the fuel oil strainer is clean.
5. Check to make sure that the fuel oil supply/storage tank has an adequate supply of fuel oil in it.
6. Check motor rotation on the fuel oil pump. Make certain that the fuel oil supply to the pump does not exceed the maximum pump inlet pressure.
7. Make sure that there is no shutoff valve in the fuel oil supply return line.
8. Bleed off any air that may have accumulated in the electric oil preheater.
9. Ensure that the manual shutoff valve in the "tracer line" is open. (Heavy Oil Only)
10. Make sure that the drain valve in the fuel oil supply line to the burner oil gun is closed (drain also used for gauge fitting to check burner oil gun nozzle pressure).
11. Make sure that the connections of the fuel oil supply and atomizing air hoses (or lines) inside the plenum box are tight.
12. Check motor rotation of the atomizing air compressor. Check the oil level in the air compressor.

Note: If shop air (plant air) is to be used for atomizing air (rather than an air compressor supplied with the burner or boiler), make sure that the air supply is adequate for the burner need. Make sure there is a manual shutoff cock and pressure gauge in the shop air supply line, upstream of the pressure regulator. A needle valve must also be installed to trim the regulated shop air supply, along with a gauge to indicate the regulated (and trimmed) atomizing air. Refer to Fig 7.2.

Note: In some instances, both an air compressor furnished with the boiler and shop air (plant air) may be available for atomizing air. If so, the source of the atomizing air must be controlled. This is accomplished with manual valves or solenoid valves and a control panel selector switch. The motor of some air compressors furnished with the boiler or burner also drives the burner fuel oil pump and motor must be provided and used when shop air is supplying the atomizing air to the burner or the drive belts to the air compressor must be removed.

Section 9:

WARNING:

Do not attempt to start up a boiler unless you have read and understand this entire manual. Only qualified and experienced boiler, burner, and controls start-up and service personnel should attempt to start any boiler.

Startup Procedure: Combination Fuels

In addition to the startup procedure for the fuel to be fired, startup for combination fuel burners require the following:

1. Set the fuel selector switch on the boiler control panel to the desired position.
2. Ensure that the register adjustment arms, located on the side of the plenum box, are adjusted to the correct position for the fuel to be burned.
3. Turn the burner operating mode switch (manual/auto) to the manual mode.
4. Turn the firing rate control dial switch (potentiometer) to its lowest setting (low fire).
5. Review the startup procedure in the following pages of this manual for the specific fuel (gas or oil) to be burned.
6. Turn the burner switch on.

With all the recycling limits in the boiler circuitry made, the load demand light on the boiler control panel will light. The boiler pre-purge will now occur. The burner modulation (MOD) motor and linkage will drive the combustion air damper and from its low fire start position to high fire, and back to its low fire start position. During this process, which takes about two minutes, more than four air changes will take place in the boiler fireside. With the low fire position proven, the burner pilot will attempt to light. After the pilot has been proven, the main fuel valve will open, the vent valve between the main fuel valves will close and the main burner will attempt to light (for gas). After the main burner flame has been established, and proven, the burner pilot will shut off. The boiler and burner operation is now under the command of the boiler and burner programmer.

7. Check the low fire hold control (aquastat), located in the forward lower side of the boiler pressure vessel, for the proper setting. The minimum setting should be 180°F.
8. Set the burner operating mode switch (manual/auto) to the "auto" position.
9. Turn the firing rate control dial switch (potentiometer) to its maximum setting.

The burner is now capable of driving from low fire to high fire, and anywhere in between, as the demand may require, but only after the low fire hold control (aquastat) has been satisfied (minimum boiler water temperature of 180°F).

Section 9:

STARTUP / SERVICE REPORT

The following document should be completed by an authorized service technician at startup and after any adjustments are made. A copy of this document must be submitted to Johnston Boiler Company to validate the warranty.

Model Number _____ Serial Number _____ Startup Date _____

	Gas			Oil		
	Low Fire	Mid Fire	High Fire	Low Fire	Mid Fire	High Fire
Stack %O ₂						
Stack CO ₂ PPM						
Stack %CO ₂						
Stack NO _x PPM						
Firing Rate (CFH/GPH)						
Main Flame Signal						
% Combustion Efficiency						
Blower Wheel Inlet Pressure ("WC)						
Plenum Box Pressure ("WC)						
Rear Combustion Chamber Pressure ("WC)						
Stack Outlet Pressure ("WC)						
Fuel Supply Pressure Before Main Regulator (psi)						
Regulated Fuel Pressure (After Main Regulator) (psi)						
Manifold Gas Pressure ("WC)						
Oil Pressure at Nozzles (psi)						
Atomizing Air Pressure at Nozzles (psi)						
Ambient Temperature (°F)						
Stack Temperature (°F)						

Service Technician Signature

Printed Name

Date

Section 9:

B. Adjustments

General

When possible, all Johnston burners are fire tested and adjusted before shipment by qualified burner personnel. However, due to combustion-specific variations such as fuel quality, air composition and density, and operating conditions, additional adjustments at the jobsite are often necessary.

The fine tuning of a burner requires a proper fuel air ratio. A proper fuel air ratio is achieved by adjusting both the air flow and the fuel flow. Proper fuel air ratio is tested using the make-up of the stack gases and the stack temperature.

Proper stack gases should contain from 3 to 3.5% O₂, less than 50PPM CO (90PPM for lo-NOX burners), and 30PPM or less NOX for lo-NOX burners. Consult factory for proper stack temperature based on your boiler model.

Swirler Vane Adjustment

The vanes of the swirler can be adjusted to allow the correct amount of air through them. Adjusting the vanes will also adjust the amount of swirl inflected upon the air that travels through them.

Controlling Amount of Fuel Flow

The control systems used when setting up the burner are designed to control the amount of fuel that enters the burner. The most common system installed on boilers is the single point positioning system. It employs a single modulation motor that is connected to a shaft that controls the opening and closing of the fuel control valve(s) and air damper simultaneously. There are three ways to adjust the fuel flow on a single point positioning system:

1) Adjust linkage joint position. This adjustment has the largest effect on firing rates. Move the joint between the linkage rod and the linkage arm to a different hole on the linkage arm. Doing so will change the linkage arm's radius. By moving the joint towards the pivot of the linkage arm, a driven linkage arm will increase rotation greatly. By moving the joint away from the pivot of the linkage arm, a driven linkage arm will decrease rotation. If the joint location is moved towards the pivot of a driving arm, the rotation of the driving arm will not be affected, but the linkage rod will move slower, causing the driven arm to rotate slower. If the joint is moved away from the pivot of a driving arm, the driven arm will move quicker.

2) Change the stroke or length of the linkage rods. This will change how fast or how slow the modulating firing rate motor changes the fuel or air settings. By increasing the amount of "dead travel" that a linkage arm must go through, the firing rate will be slowed down. By decreasing the amount of "dead travel", the firing rate is sped up, and the burner will travel to high fire more quickly.

3) Adjust the fuel flow through the characterized fuel control valve. This is done by matching the fuel flow with the air flow at each point through its travel range. Consult fuel control valve manual for adjustment instructions.

A parallel point positioning system allows the fuel and air to be introduced at varying rates. This is the more modern method of fuel air ratio control and requires a modulation motor for each fuel control valve and one for the air damper. These systems are computer controlled and require some experience to operate. Only trained personnel should attempt to adjust or modify these settings.

The optimization of the fuel air ratio is essential to efficient operation of any burner. Analysis of flue gases with proper instruments is the only way of truly optimizing fuel air ratio.

Johnston Boiler Company recommends flue gas oxygen levels of around 8% in low fire and 3-4% in high fire for gas fired burners. Oxygen level recommendations for oil fired burners are around 6% in low fire and 4-4.5% in high fire.

Section 9:

Johnston Boiler Company recommends no greater level than 200 ppm of Carbon Monoxide for both low fire and high fire. The acceptable industry standard level is no more than 200 to 400 ppm of Carbon Monoxide. Carbon Monoxide is an indication of incomplete combustion and/or poor mixing.

Johnston Boiler Company also recommends no combustibles be present in the flue gases. Combustibles are a good indication of incomplete combustion.

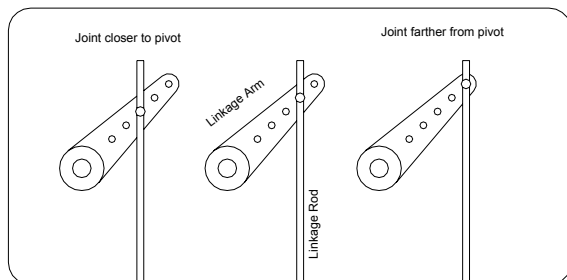


Figure 9.1: Linkage Joint Adjustment

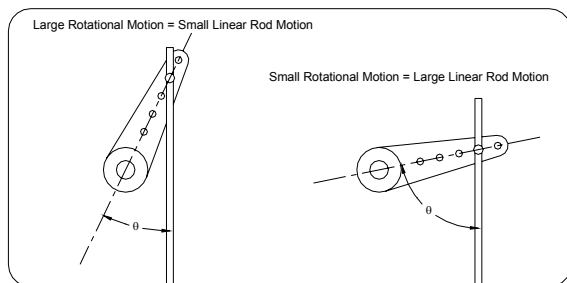


Figure 9.2: Linkage Arm Modification

If oxygen levels are too high, too much excess air is being allowed into the combustion chamber. High oxygen levels may also cause loud flame vibrations. To remedy this, adjust the inlet air damper to allow less air in or adjust the fuel control valve to allow more fuel to the flame front.

If oxygen levels are too low, not enough air is being let into the combustion chamber. Open the inlet air damper to allow more air in or tune the control valve to allow less fuel to the flame front.

High Carbon Monoxide levels are an indication of either too much oxygen or not enough oxygen is being introduced to the flame front. This must be adjusted using the fuel control valve and the inlet air damper.

With gas burners, at high fire, with a proper fuel air ratio, the flame should appear blue. With a Johnston Lo-NO_x FGR system, the flame should appear dark blue or even a purple color at high fire.

At high fire on an oil burner, the flame should appear bright yellow if a proper fuel to air ratio is achieved.

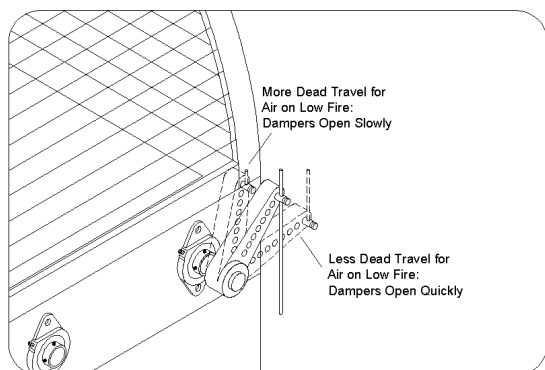


Figure 9.3: Inlet Air Damper Adjustment

Section 9:

Inlet Air Damper

On burners equipped with an inlet air damper, the airflow through the burner is limited by the position of the inlet air damper blades. If the air blades are too far closed during high fire, the burner will be under-fired even if properly tuned. A general rule is that the air blades should be at least 80% open when on high fire.

On Johnston "S" Burners, the inlet air damper is between the silencer and the air inlet housing. Fig 6.1

The inlet air damper controls the amount of air let into the blower. For burners with a parallel positioning system, the air damper is controlled directly by the burner programmer which in turn control individual motors located directly on the damper shafts (see Fig 6.1).

For burners with a single point positioning system, linkages from the MOD motor are used to control the inlet air damper. These linkages may be adjusted as follows:

- To decrease the amount of air let into the burner, increase or decrease the length of the linkage rod to obtain a larger angle between the rod and the linkage arm. This will decrease the amount of rotational travel for an equal amount of travel made by the arm.
- To increase the amount of air let into the burner, increase or decrease the length of the linkage rod to obtain a smaller angle between the rod and the linkage arm. This will increase the amount of rotational travel for an equal amount of travel made by the arm.

Firing Rate Modification

Firing rate is the rate at which fuel is burned. Firing rate is controlled on gas and oil burners by the modulating control valves.

For burners equipped with a parallel point positioning control system, firing rate can be adjusted directly from the boiler programmer. See programmer manual for instructions on adjustments.

For burners equipped with a single point positioning system, as seen in figure 6.2, adjustments are made by modifying the linkages from the MOD motor.

The firing rate can be adjusted on gas burners by adjusting the linkage connection to the control valve. To increase the firing rate, lengthen the connecting rod to create a great angle between the rod and the linkage. To decrease the firing rate, shorten the connecting rod to decrease the angle.

The firing rate on oil burners is modified by adjusting the linkage to the oil control valve. This valve is adjusted just like the gas control valve.

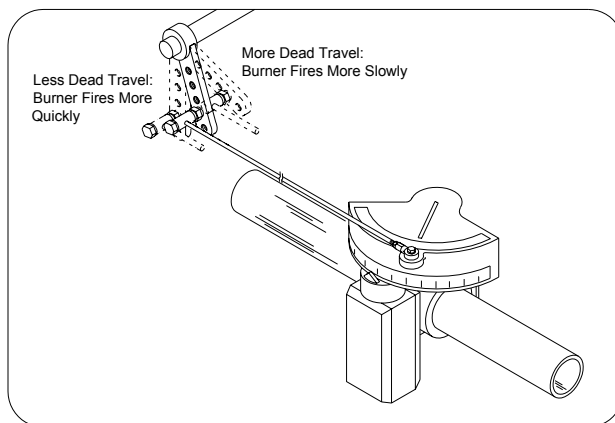


Figure 9.4: Fuel Valve Modification

Section 9:

FGR Percentage

Increasing the amount of flue gases that are recycled on an FGR equipped burner will decrease the amount of NO_x emissions present in the stack flue gas. It will also, however decrease the amount of oxygen present in the air entering the combustion chamber.

Therefore, a balance must be reached that minimizes NO_x emissions while still allowing efficient firing. This can be done by adjusting the FGR inlet damper and monitoring the stack O₂ and CO to measure firing efficiency.

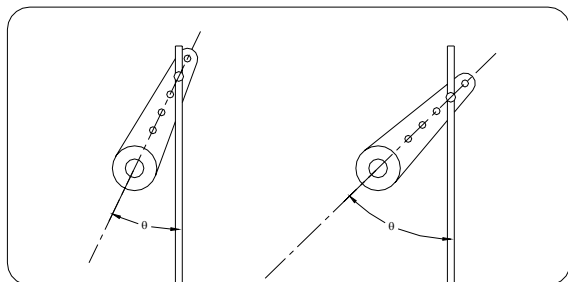


Figure 9.5: Linkage Modification

To adjust the FGR percentage, first bring the boiler to low fire. For burners equipped with a parallel positioning control system, FGR percentage is modified from the burner programmer. See programmer manual for information on adjustments. For burners equipped with a single point positioning control system, the linkages to the FGR damper must be modified. Increasing the angle θ as shown in figure 5.5 will increase the amount of FGR introduced off of low fire. There is also a fixed shutoff damper in the FGR line. It can be used to control the maximum amount of FGR introduced.

Gas Ring Adjustment

If the burner is not firing correctly on gas after adjusting the fuel air ratio, the distribution of gas to the inside of the flame may not be adequate. Getting more gas to the inside of the flame is achieved by orifacing the gas spuds. Consult the factory for proper orifice sizes to be used on your burner.

Oil Gun Location

The location of the oil gun relative to the diffuser is important when firing on oil. If the burner cannot be tuned by adjusting the fuel air ratio, check the location of the oil gun tip. The tip of the oil gun should generally be between 0.00 and 2.00 inches from the diffuser. To adjust this, start with the oil gun flush with the diffuser and move it back until the burner is adequately tuned. If the tip of the oil gun is placed more than 2 inches from the diffuser, oil could spray on the diffuser, which is to be avoided.

Section 9:

C. Shutdown

Shutdown Procedures

Within this manual, two shutdown conditions are defined as follows:

Normal Shutdown: The flame is extinguished by either the control system (automatic) or the user (manual) with the intent that it be restarted within a short time period. Such action is not typically associated with a problem and, unless otherwise indicated by the control system, is not a cause for alarm.

Emergency Shutdown: The flame is extinguished by the control system (automatic) or the user (manual) and is the result of a system problem or equipment malfunction. Under such conditions, the burner may not be restarted until the problem has been corrected and the appropriate safety device(s) have been reset. The manual emergency shutdown procedure should also be followed for extended idle periods where automatic restarting is not required, and at any time that maintenance is to be performed on the boiler/burner.

Normal Shutdown Procedure

Automatic (Recycling): No action by the boiler operator is required. Operating controls (pressure / temperature sensors) open automatically. The operating control senses the boiler water temperature or steam pressure and shuts down the burner when it has reached the setpoint. This is the most common shut down and is a normal part of the burner operation. When the water temperature or steam pressure drops, the system will automatically restart the burner.

Manual: Drive burner to low fire. Turn burner switch to "OFF" position. (Some control panels may have a "Burner Off" pushbutton which must be depressed, rather than a selector type switch)

Note:

If maintenance is to be performed on the boiler/burner, follow manual procedure for emergency shutdown as described below.

Warning:

If a starting failure occurs for any reason, combustible fuel gases may fill the furnace and combustion chamber. Never attempt to relight the burner under these conditions without first purging the unburned fuel.

Emergency Shutdown Procedure

Automatic (Non-Recycling): No action by the boiler operator is required for an automatic emergency shutdown to occur; however the manual emergency shutdown procedure must be followed after an automatic emergency shutdown. In this instance the control system extinguishes the flame automatically and will not restart until the problem is corrected. Such a shutdown is caused by unsafe conditions, such as low water, flame failure, high steam pressure or water temperature, high or low fuel pressure, etc., and is usually accompanied by an audible or visible alarm signal.

Manual:

1. Turn the burner switch to "OFF" position (or depress the "Burner Off" pushbutton if so equipped).
2. Close all manual fuel valves.
3. If equipped, turn oil pump on/off switch to off.
4. Close main blowoff or drain valves.
5. Close continuous blowdown stop valve if present at facility.
6. Close main feedwater and chemical feed inlet valves if present at facility.
7. Close hot water valves or steam stop valve(s) if present at facility.
8. Lock out incoming power according to plant specifications at main disconnect panel (necessary if work is to be performed on unit).
9. Do not attempt to reverse these actions or to restart the burner until the problem(s) has been identified and corrected.